

1. DEFINITION

DEFINITION
Propylene glycol dicaprylocaprate EP / Propylene glycol dicaprylate/dicaprate NF

DENOMINATION	
INCI NAME (PCPC - International Nomenclature of Cosmetic Ingredients)	Propylene Glycol Dicaprylatedicaprate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	04446S9CRA
ADDITIONAL INFORMATION ON THE DENOMINATION	Propylene glycol dicaprylate/dicaprate [INCI] Propylene glycol dicaprylate Decanoic acid, mixed diesters with octanoic acid and propylene glycol

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Bulk Propylene glycol dicaprylocaprate is obtained by an esterification reaction between Propylene glycol and Caprylic/Capric acids (C8-10). Gattefossé SAS purchases bulk Propylene glycol dicaprylocaprate, makes filtration, packaging, control and batch release

COMPOSITION	
Well defined multi-constituent substance composed of Propylene glycol diesters of Caprylic/Capric acids (C8 and C10) (diesters > 95%)	
Full Labeling (Key letters system)	
A₁ ≥ 75.0 ; 50.0 ≤ A₂ < 75.0 ; 25.0 ≤ B < 50.0 ; 10.0 ≤ C < 25.0 ; 5.0 ≤ D < 10.0 ; 1.0 ≤ E < 5.0 ; 0.1 ≤ F < 1.0 ; G < 0.1	
Propylene Glycol Dicaprylatedicaprate	A ₁ ≥ 75.0%



Regulatory data sheet (RDS) – Pharmaceutical market

LABRAFAC PG (Code: 3230)

Last update: 05 January 2022

COMPOSITION
Other constituents (i.e. incidental ingredients)
None

4. DRUG MASTER FILE

A Drug Master File (DMF) is a submission to the Food and Drug Administration (FDA) that may be used to provide confidential detailed information about facilities, processes, or articles used in the manufacturing, processing, packaging, and storing of one or more human drugs. The submission of a DMF is not required by law or FDA regulation. A DMF is submitted solely at the discretion of the holder. (Source: US FDA)

In general, the excipient DMF is not used for compendial excipients but more for novel excipients (e.g., new excipients, mixtures, coprocessed materials, and biotech excipients).

FDA usually does not review DMFs for compendial excipients unless there is a specific reason to do so (e.g., new route of administration or application for existing excipients).

(Source: IPEC Americas)

We have no DMF for LABRAFAC PG because it is a compendial excipient.

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopeia Current edition	Propylene glycol dicaprylocaprate
USP/ NF Current edition	Propylene glycol dicaprylate/dicaprate

6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓ (9) warning: cat	✓ (1) All species (No limit)	✓ (9) warning: cat	✓ (5) USFA 172.856	✓	✓ 21CFR180.910



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LABRAFAC PG (Code: 3230)

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EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites

EUROPEAN REGULATION
 (1) Regulation (EU) No 37/2010 - Listed in annex (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption)
 (2) Regulation (EU) No 37/2010 - Listed in annex (Polyethylene glycol stearates with 8-40 oxyethylene units)
 (3) Regulation (EU) No 37/2010 - Listed in annex (Diethylene glycol monoethyl ether)
 (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin

USA REGULATION
 - Veterinary drugs
 FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety
 (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR)
 (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database)
 - Pesticides
 (8) EPA's pesticides program (<http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::>)

7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	propylene glycol dioctanoate/didecanoate Propylene glycol dicaprylate/dicaprate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines	Only for use in topical medicines for dermal application. Listed Medicines

AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.

8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE					
Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute toxicity (Dose escalating)	IV bolus (tail vein)	Mouse	LABRAFAC™ PG Vehicle: Physiological saline solution M + F: 25, 50, 100, 200, 400, 800, 1600, 6400, 3200 and 4800 mg/kg	MTD(IV): 3200 mg/kg	(Gattefossé data, GAT-MTDs-PH- 08/0473, 2009)



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SAFETY STUDIES PERFORMED BY GATTEFOSSE

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute toxicity (eq OECD 423)	IV bolus (tail vein)	Mouse	LABRAFAC™ PG Vehicle: Physiological saline solution M+F: 3200, 1600 mg/kg	MTD(IV): 1600 mg/kg Administration at 3200 mg/kg induces mortality (1/5 male and 3/5 females). Administration at 1600 mg/kg induces no mortality and toxicity signs.	(Gattefossé data, GAT-TAIs-PH- 08/043, 2009)

CIR COMPENDIUM (Cosmetic Ingredient Review) (<http://www.cir-safety.org>)

Denomination	Reference	Maximum “as used” concentration	Conclusions
Propylene glycol diesters	2015, Jan 15	Up to 51.8 %	The CIR Expert Panel concluded that Propylene Glycol Dicaprylate/Dicaprate is safe in the present practices of use and concentration described in this safety assessment in cosmetics when formulated to be nonirritating.

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION

	CAS #	EINECS #	Chemical name
Propylene Glycol Dicaprylatedicaprate	68583-51-7	271-516-3	Decanoic acid, mixed diesters with octanoic acid and propylene glycol

NATIONAL INVENTORIES

	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AICS)	China (IECSC)	Other inventories
Propylene Glycol Dicaprylatedicaprate	Listed	DSL listed	Listed	Listed	TSCA, DSL, EINECS, AICS, PICCS, ASIA- PAC, NZIoC, ECL Serial No. KE-09464, IECSC, TCSI

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics ... are not subject to TSCA regulation.
Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. In addition, substances not listed in DSL or NDSL, but listed in the “in-commerce list” can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as “Natural” according Health Canada’s definition are exempted from New Substances regulation.
Australia: Substances Introduced at 1% or less in cosmetics are exempted from AICS listing. Otherwise, substances not listed on AICS can be imported in Australia in quantities not exceeding 10 kg per 12-month period and per importer. In addition, in the Industrial Chemicals (Notification and Assessment) Act 1989 (the Act), an industrial chemical which meets the definition of ‘a naturally-occurring chemical’ is taken to be on the Australian Inventory of Chemical Substances (AICS), whether listed or not, and therefore does not require notification as a new chemical prior to manufacture or importation into Australia.



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LABRAFAC PG (Code: 3230)

Last update: 05 January 2022

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	Status per substance
Propylene Glycol Dicaprylatedicaprate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.

10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)		
Inactive ingredient	Route / Dosage form	Use level
No FDA IID reference exactly matches the definition Propylene Glycol Dicaprylatedicaprate		
Maximum potency: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the "maximum potency" field will be blank.		

10.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)		
Inactive ingredient	Route / Dosage form	Use level
Propylene glycol dicaprylate (UNII : 581437HWX2)	TOPICAL (Emulsion, cream)	10 % w/w



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FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)		
Inactive ingredient	Route / Dosage form	Use level
Propylene glycol monocaprylate (UNII: RT9P9S09QI)	ORAL	Tablet, film coated
Maximum potency: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the "maximum potency" field will be blank.		

10.2. Food/Nutraceutical/Dietary supplement applications

Propylene glycol esters of Fatty acids have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS 477 Propane-1,2-diol esters of fatty acids
E.U.	EFA (European Food additives)	E 477 Propane-1,2-diol esters of fatty acids
U.S.A.	DSHEA (Dietary Supplement Health and Education Act) "Grandfathered ingredient" authorized for use in Dietary supplements	Esters of propylene glycol
U.S.A.	FCC (Food Chemical Codex)	Propylene glycol mono- and diesters
U.S.A.	USFA US Food additives according to 21 CFR	21CFR §172.856 Propylene glycol mono-and diesters of Fats and Fatty acids
JAPAN	JSFA (Japanese Standards of Food Additives)	Propylene glycol esters of Fatty acids

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

Liability clause

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